Arghyadeep Deb

AI/ML Engineer | Bhubaneswar, India

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Professional Summary

Computer Science undergraduate specializing in **AI/ML and Agentic AI**, with hands-on experience building autonomous multi-agent systems, RAG applications, and production-oriented ML solutions. Experienced in **LLM orchestration, LangGraph, MCP, machine learning, deep learning, and NLP**, with a track record of developing end-to-end AI pipelines and scalable APIs using **PyTorch and FastAPI**.

Education

B.Tech in Computer Science, Kalinga Institute of Industrial Technology (KIIT) (CGPA: **8.55**) **2023 – 2027**

Skills

Languages: Python, Java, SQL, JavaScript, HTML, CSS
AI/ML: Machine Learning, Deep Learning, NLP, Generative AI, RAG, Agentic AI, Multi-Agent Systems, SHAP
Frameworks: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, FastAPI
AI Infrastructure: MCP, LangSmith, Vector Databases, Ollama
Tools & Platforms: Git, GitHub, Power BI, Docker, Claude Code, Codex

Professional Experience

- **AI Intern — Cygnatrix IT Solutions** **May 2026 – Jun 2026**
  Designed and built the **AI layer and FastAPI backend** for **Grove**, a thinking-to-knowledge platform. Orchestrated **3 LangGraph pipelines** and a **LiteLLM** provider layer with fallback chains, plus **LightRAG** retrieval, validated by **211 backend tests**.
- **Intern — State Bank of India (SBI)** **May 2025 – Jun 2025**
  Ran a survey-based research project on **"AI in Retail Lending: Enhancing Customer Insights for Housing and Personal Loans."** Surveyed 150 respondents in Guwahati and authored a report on how the bank could adopt AI.
- **ML Intern — Intrainz** **Jul 2025 – Sep 2025**
  Built and evaluated ML pipelines on real-world datasets using **EDA, PCA, and regression modeling**, with **tuning and cross-validation**.

Projects

- **AI Financial Advisor (Agent-Based System)**
  Built a **multi-agent advisory platform** with **7 specialized agents** and **16 FastAPI endpoints**, with SQLite persistence and a RAG + GraphRAG knowledge base. Deployed on **Render + Streamlit**.
- **Fairness-Aware Credit Scoring (B.Tech Major Project)**
  Trained a **TabTransformer** with a joint fairness loss on Home Credit data; gender is never an input. Prior correction cut worst-group ECE from **0.35 to 0.013**, and a fairness step cut the equalized-odds gap from **0.060 to 0.015** at unchanged AUC-ROC (0.746).
- **Hybrid Movie Recommender System**
  Built an item-to-item **hybrid recommender** blending TF-IDF content similarity with SVD-based collaborative similarity trained on **22.6M ratings**, with a tunable blend weight, served via **FastAPI** with a Streamlit demo.

Additional Information

- **Event Coordinator — KIZUNA, KIIT's Japanese Cultural Society** **2026**
  Organized **KAIZEN 1.0**, a two-day Japanese cultural fest (300–500 attendees), leading promotion and writing the festival content.
- **Founder — Social Media Marketing Agency** **2024 – 2025**
  Ran an independent agency for restaurant clients, owning the full client lifecycle from outreach to delivery.
- Languages: English, Hindi, Bengali, Assamese, Japanese (N5, learning)

Certifications

- Introduction to Deep Learning & Neural Networks with Keras — IBM **2026**
- Generative AI Essentials — IBM **2026**
- Data Analysis with Python — IBM **2026**
- AI Foundations & Generative AI — Oracle **2025**
- Data Science Professional Certificate — Oracle **2024**